Personalising circadian hygiene educational initiatives aimed at university students: "He who has ears to hear, let him hear"

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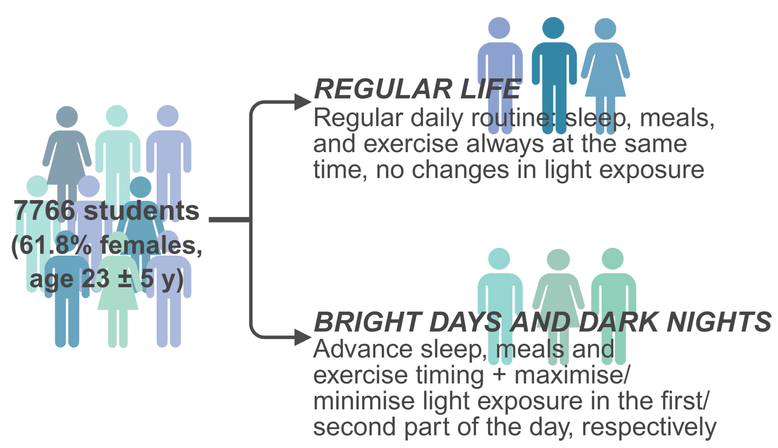
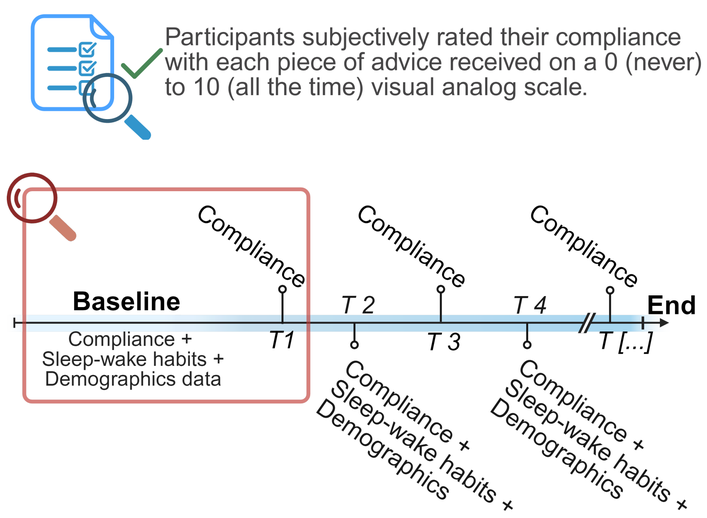
UNIVERSITY OF SURREY

INTRODUCTION

- Health-related surveys and educational initiatives, particularly online, are prone to different types of selection bias (e.g., personal interest, computer/social media literacy; (Ebert J et al., 2018. J Med Internet Res).
- Without detailed knowledge of the target population's characteristics, evaluating participation and attrition *post hoc* can be challenging (Wu MJ et al., 2022. Computers in Human Behavior Reports).
- A group of "early drop-outs" of the University of Padova SleepRhythm Unipd circadian education initiative were utilised to assess its efficacy and guide its development (Montagnese S et al., 2022. FrontNeurosci).
- Launched in October 2019, the initiative aimed at improving sleep timing and quality through one of two sets of circadian hygiene advice ("A regular life" or "Bright days and dark nights"), covering sleep timing, meal timing, physical exercise timing, and light exposure. They were also asked how easy it had been to comply with the advice.

METHODS





Electronic versions of the questionnaire were used:

- Pittsburgh Sleep Quality Index (PSQI)
- Sleep Timing Sleep Quality Screening (STSQS) questionnaire
- Ultrashort Version of the Munich ChronoType Questionnaire mMCTQ
- Epworth Sleepiness Scale (ESS)
- Self-Morningness/Eveningness Self-ME Question

CONCLUSION

- High prevalence of poor night sleep and social jetlag among University of Padova students, particularly those with evening or extremely evening chronotype
- Evening-type students showed lower compliance and higher drop-out rates

RESULTS

- 4640 (59.7%) students had abnormal night sleep quality (PSQI score >5), 967 (12.4%) had excessive daytime sleepiness (ESS ≥11), and average social jet lag was 1.2±1.2 hr.
- 3868 (49.8%) students were assigned to the "Bright days and dark nights", 3898 (50.2%) to the "A regular life" advice groups, and there were no significant differences between the two groups.
- 3185 students (41.0%, "early drop outs") did not complete follow-up questionnaires, depending on the study entrance date, they had a period of time to do so between 2 and 43 months.
- The likelihood of never providing follow-up data was higher in males (46% versus 37%; χ^2=58, p<0.0001).
- Higher proportion of evening types among males (Table 1) and a higher proportion of extremely evening/evening types amongst the early drop-outs (χ^2=10, p<0.05).
- 3035 (39%) students completed the first compliance questionnaire; males and extremely evening/evening types were significantly less prevalent compared with the baseline sample (χ^2=8, p<0.01; χ^2=19, p<0.001, respectively).
- Students who did not provide follow-up information had worse sleep-wake habits than those who did (Table 2).

	Extremely morning	Morning	Evening	Extremely evening	Total (n)
Males (n, %)	250 (8.4%)	934 (31.5%)	1181 (39.8%)	604 (20.3%)	2969
Females (n, %)	557 (11.6%)	1602 (33.4%)	1833 (38.2%)	805 (16.8%)	4797
Total (n)	807	2536	3014	1409	7766

Table 1 Chronotype distribution, by sex. Significant difference with a higher proportion of evening types among males was shown. *Note: χ^2 = 34, p < 0.0001.*

	Follow-up *n* = 4581	No follow-up *n* = 3185
Age (years)	23.2 ± 5.4	23.3 ± 5.1
Mood (1–10)	5.9 ± 1.9	5.8 ± 1.8
ESS (total score, 0–24)	6.2 ± 3.5	6.4 ± 3.6**
PSQI (total score, 0–21)	6.7 ± 3.2	6.8 ± 3.1*
Sleep-onset time (work/study days; clock time)	00:06 ± 01:12	00:12 ± 01:18***
Wake up time (work/study days; clock time)	07:30 ± 01:18	07:42 ± 01:36***
Midsleep (work/study days; clock time)	03:48 ± 01:12	03:54 ± 01:18***
Sleep duration (work/study days; hr)	7.4 ± 1.3	7.4 ± 1.4
Sleep-onset time (free days; clock time)	00:54 ± 01:18	01:00 ± 01:30***
Wake up time (free days; clock time)	09:12 ± 01:30	09:18 ± 01:18*
Midsleep time (free days; clock time)	05:00 ± 01:12	05:12 ± 01:24***
Sleep duration (free days; hr)	8.4 ± 1.3	8.3 ± 1.4
Social jet lag (hr)	1.2 ± 1.1	1.2 ± 1.3

Table 2 Age, mood and sleep–wake features of students who did/did not engage in follow-up. *Note: *p < 0.05; **p < 0.01; ***p < 0.0001 (Bonferroni-corrected for sleep–wake timing variables). Abbreviations: ESS, Epworth Sleepiness Scale; PSQI, Pittsburgh Sleep Quality Index.*

- Compliance with sleep timing advice was significantly lower in the "Bright days and dark nights" arm (4.4±2.4 vs. 5.6±2.4, p<0.001).
- Evening or extremely evening chronotype were less likely to follow the recommendation, except for sleep timing, compared to those with morning or extremely morning types (Figure 1).
- Distance learning combined with chronotype affected work and study days, but only chronotype affected free days (Figure 2).

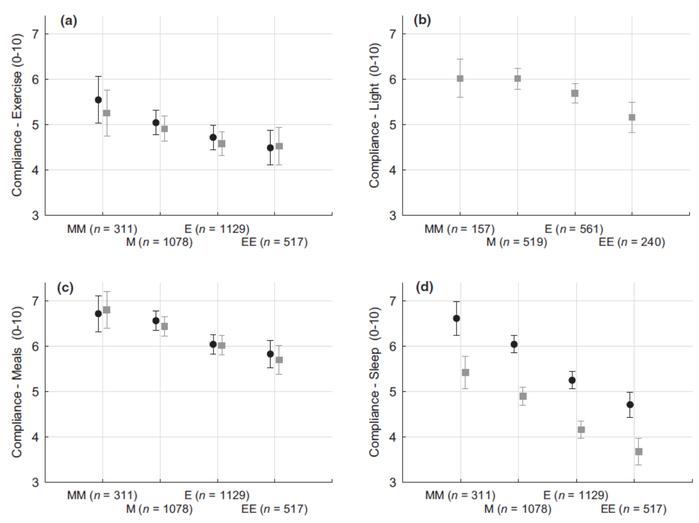


Figure 1 Self-reported compliance (1–10 visual-analogue scale, study time T1, that is, 1 month after joining; mean ± 95% CI) to different types of advice (advice on exercise [a], light [b], meals [c], and sleep [d]) in the "Bright days and dark nights" (grey squares and "A regular life" (black circles) groups, by chronotype (on the x-axis).

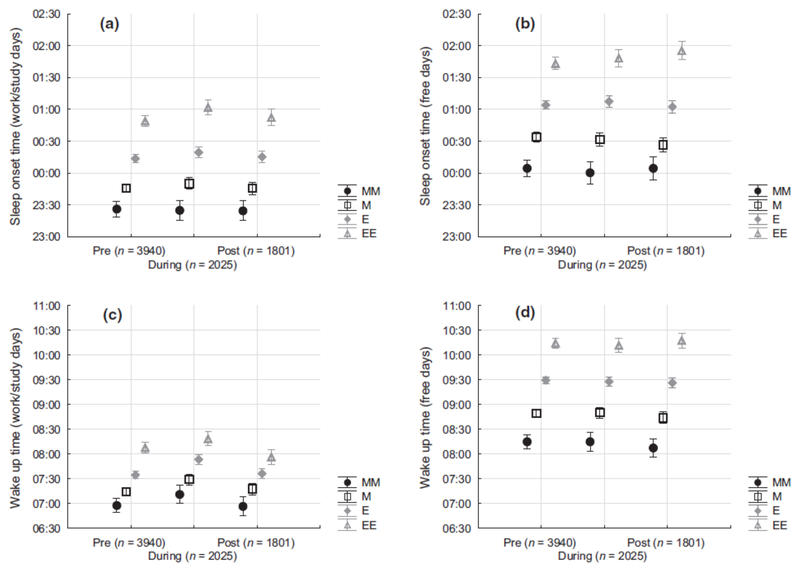


Figure 2 Sleep-onset (mean ± 95% CI; top panels, a, b) and wake up times (mean ± 95% CI; bottom panels, c, d) on work/study days (left panels, a, c) and free days (right panels, b, d), by chronotype (extremely morning: black circles; morning: empty squares; evening: grey diamonds; extremely evening: empty triangles) and distance learning (on the x-axis).

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